

Finite Communication and the Nonmonotone Value of Network Mixing^{*}

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Abstract

Many social learning environments communicate assessments through finite categories rather than continuous scores: ratings, grades, risk tiers, classifications, and platform labels. This paper studies how finite communication changes the value of network mixing. Agents carry real-valued latent assessment scores, but, under a social network, they can only observe other agents' finite messages generated by a common codebook. Under continuous communication as a benchmark, every agent aggregates social information and learns the true state. In contrast, under finite communication, information aggregation depends only on categorized messages, not on the latent scores. Movements of latent scores inside a category bin remain socially invisible, and new social information appears only when scores cross categories. Faster mixing under this finite communication therefore has two effects: it compresses disagreement, but it can also compress agents into a common message region before the state-aligned message has become socially visible. We characterize when disagreement compression becomes a same-message trap and then study which message is selected. In a canonical block network environment, increasing cross-group exposure moves the selected outcome from state-aligned consensus to non-state-aligned consensus. Accuracy is therefore nonmonotone in mixing: moderate integration can improve learning, while excessive integration can prematurely select a less informative message.

JEL Classifications: D83, D85

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1 Introduction

Many social learning environments communicate information through finite categories rather than through exact beliefs or latent scores. Credit ratings, grades, risk tiers, classifications, platform labels, and review scores all compress richer underlying assessments into standardized messages. In such settings, information can move inside a category while the public message remains unchanged. A score may become more favorable, a risk assessment may improve, or a product evaluation may strengthen, but these changes are not socially visible until the score crosses into another category. This paper studies how this finite communication layer changes the value of network mixing.

The standard intuition is that more mixing improves social learning. When agents are exposed to a broader set of others, private information travels farther, local disagreement is averaged out, and society becomes less segmented. This force is central to linear social-learning models in the DeGroot tradition and to work on network learning in large societies (DeGroot, 1974; Golub and Jackson, 2010). We keep the basic force of social averaging but change what is averaged. Agents carry real-valued latent assessment scores and update them over time using private evidence and social information from a network. However, they observe others only through finite messages generated by a common codebook. Social aggregation is therefore an average of finite messages, not an average of latent scores.

This distinction creates the central friction. In each period, agents combine socially observed messages with private evidence and update their latent scores. The resulting update becomes socially visible only when a score crosses into a different category. Movements within the same category are therefore privately carried but socially invisible. Finite communication consequently turns learning from continuous evidence accumulation into path-dependent selection among message regions. Network mixing still compresses disagreement, but the same compression can make a common message region absorbing before the state-aligned message has become socially visible.

We first show that the updating rule itself does not create the learning trap or failure. In a continuous-communication benchmark, agents transmit their latent scores directly. Under the paper’s score-updating rule, every agent learns the true state. Thus, the obstacle is not private evidence, nor social averaging per se. The failure comes from the finite communication. Once agents communicate the decoded categories rather than scores, the network no longer observes within-bin evidence accumulation.

We then study when finite communication creates same-message traps. The mechanism is straightforward. Network mixing makes agents’ assessments more similar. With continuous

communication, this helps information spread. With finite communication, however, it can push many agents into the same category before their latent scores have moved far enough. Once many agents send the same message, that message can become self-reinforcing: agents keep observing the same category.

The general trap result explains persistence, but not selection. It shows when a large same-message group remains stable, but it does not determine which message the society selects. To study selection, we use a simple block environment with two groups and three messages: low, interior, and high. The high message is a state-aligned message. One group initially sends the high message, while the other group initially sends the low message. When cross-group mixing is low, the groups remain segmented. With moderate mixing, the high-message group pulls the low-message group upward, and society reaches high-message consensus. With too much mixing, the two groups average each other too quickly, and both move into the interior message instead.

The key implication is that more mixing need not always improve accuracy. Moderate integration can help because the state-aligned group spreads the correct message. Excessive integration can hurt because it selects a less informative interior message before the high message has time to spread. The problem is therefore not integration itself. The problem is premature integration when communication is finite.

This gives a timing view of communication design. More exposure is valuable after informative assessments have crossed public thresholds, because the right message can then travel through the network. The same exposure can be harmful before those crossings occur, because it can compress the society into a less informative category. In settings organized around ratings, grades, risk tiers, classifications, or platform labels, the design question is not simply whether to increase mixing. It is when to increase mixing, given that finite messages reveal only category changes and hide movements within categories.

1.1 Related Literature

This paper contributes to the literature on social learning in networks. The continuous-communication benchmark is closest to the DeGroot tradition, in which agents repeatedly average the opinions or beliefs of their neighbors (DeGroot, 1974). Subsequent work studies how repeated averaging, social influence, and non-Bayesian updating aggregate information in networks (DeMarzo et al., 2003; Golub and Jackson, 2010; Jadbabaie et al., 2012; Molavi et al., 2018). The present paper keeps the repeated-averaging structure but adds a finite communication layer. Agents carry continuous latent scores, while the network observes only decoded finite messages. This distinction changes the role of mixing. In the continuous

benchmark, irreducible averaging transmits accumulated private evidence and learning succeeds. Under finite communication, the same averaging rule operates on coarse messages, so within-bin private learning is not socially transmitted.

The paper also relates to Bayesian and observational learning models in which agents infer information from the actions of others. Classic models of informational cascades and herd behavior show that rational agents may eventually ignore private information after observing a sufficiently informative history of past actions (Banerjee, 1992; Bikhchandani et al., 1992). Later work studies pathological outcomes and network restrictions on observational learning (Acemoglu et al., 2011; Smith and Sørensen, 2000). The mechanism here is different. Agents continue to receive private evidence, and the failure is not caused by optimal inference from a public action history. Instead, private evidence affects the social environment only when it moves a latent score crossing a category. Absorption arises because finite messages make within-bin movements invisible to others.

The finite-message structure connects the paper to work on categories, certification, and coarse signal communication. In information design and certification settings, categories can shape what information is revealed and how receivers interpret underlying value (Lizzeri, 1999; Onuchic and Ray, 2023). In social learning, Xu (2025) studies learning from coarse signals of others' actions. There is also a related literature on quantized consensus and granular DeGroot dynamics, where communication or opinions are restricted to a finite grid (Amir et al., 2025; Kashyap et al., 2007). The present paper differs in the object being discretized and in the main comparative static. Agents update continuous latent scores, but they transmit only categorical messages. The key result is not simply convergence failure under coarse communication, but nonmonotone accuracy in network mixing: more mixing can improve learning after informative threshold crossings occur, yet reduce accuracy when it compresses disagreement before those crossings become socially visible.

The block-mixing analysis is related to work on homophily, segregation, and block-structured networks. Stochastic block models provide a standard representation of group-based network structure (Holland et al., 1983), and economic models of homophily explain why social networks often display within-group concentration of ties (Currarini et al., 2009). In learning models with repeated averaging, homophily typically slows convergence by preserving group-level disagreement (Golub and Jackson, 2012). The present paper shows that, with finite communication, the welfare effect of integration is more subtle. Low mixing can preserve segmentation, moderate mixing can select state-aligned consensus, and excessive mixing can prematurely select an interior message. Thus the issue is not only how fast disagreement disappears, but whether the relevant information has become publicly visible before dis-

agreement is compressed.

The paper proceeds as follows. Section 2 introduces latent scores, finite codebooks, social exposure, private evidence, and the accuracy criterion. Section 3 studies the continuous-communication benchmark and shows that the same updating rule learns when latent scores are transmitted directly. Section 4 characterizes how finite communication turns fast mixing into same-message absorption and persistent accuracy loss. Section 5 studies the canonical block environment and derives the phase diagram in which low, moderate, and excessive mixing select different absorbing message regimes. Section 6 concludes.

2 Model

Agents carry real-valued assessment scores but communicate through finite messages. These two objects should be kept separate. A score is an internal assessment score. A message is the finite category generated from that score and observed by others. The paper’s friction is that the network observes the message, not the score.

This section introduces the environment in the order in which these objects operate. We first define the state, agents, and latent assessment scores. We then define the finite codebook and the network aggregate of decoded values. After that, we introduce private evidence, the score-updating rule, timing, and the accuracy criterion.

2.1 State, agents, and latent scores

Time is discrete, $t = 0, 1, 2, \dots$. There is a binary state $\theta \in \{0, 1\}$, drawn at date 0 with $\Pr(\theta = 1) = \mu_0 \in (0, 1)$ and fixed thereafter. The probability μ_0 describes the ex ante distribution of the state. For each society size $n \geq 2$, the set of agents is $N = \{1, \dots, n\}$.

Agent i ’s private state variable at date t is a real-valued latent score $\ell_{i,t} \in \mathbb{R}$. Higher scores represent stronger assessments in favor of $\theta = 1$, while lower scores represent stronger assessments in favor of $\theta = 0$. The score is an assessment index, not necessarily a Bayesian log posterior odds. The probability interpretation used for evaluating accuracy is introduced below through the logistic map.

The initial score profile

$$\ell_0 = (\ell_{1,0}, \dots, \ell_{n,0}) \in \mathbb{R}^n$$

is deterministic. We allow heterogeneous initial scores. Initial scores are uniformly bounded:

$$\max_{i \in N} |\ell_{i,0}| < \infty.$$

The initial scores are fixed starting assessments at the beginning of the communication process.¹ Agents know only their own initial scores.

2.2 Finite communication and social exposure

Agents do not transmit latent scores directly. Instead, communication is described by a common finite codebook. The codebook consists of a finite message set, an encoder, and a decoder. The encoder maps each score into one of finitely many messages, and the decoder assigns a score-scale value to each message. This captures environments in which communication is standardized, such as ratings, grades, risk tiers, classifications, or platform labels. The same message has the same decoded meaning for all receivers, and agents do not strategically choose messages to influence others.²

Fix $K \geq 2$, and let $\mathcal{M} = \{1, \dots, K\}$ be the message set. An encoder is a measurable map $Q : \mathbb{R} \rightarrow \mathcal{M}$, and a decoder is a map $\Psi : \mathcal{M} \rightarrow \mathbb{R}$. The codebook $(\mathcal{M}, Q, \Psi)$ is common across agents. If agent j has score $\ell_{j,t}$ at the communication stage of date t , she sends

$$m_{j,t} = Q(\ell_{j,t}),$$

and receivers assign the decoded value

$$y_{j,t} = \Psi(m_{j,t}).$$

Equivalently, $y_{j,t} = \psi(\ell_{j,t})$, where $\psi = \Psi \circ Q$.

The main text focuses on threshold codebooks. Let

$$-\infty = b_0 < b_1 < \dots < b_{K-1} < b_K = +\infty,$$

and define $B_k = (b_{k-1}, b_k]$ for $k = 1, \dots, K$. The encoder assigns message k whenever $\ell \in B_k$, and the decoder assigns the score-scale reconstruction $r_k = \Psi(k)$. Thus

$$\psi(\ell) = r_k \quad \text{for every } \ell \in B_k, \quad k = 1, \dots, K.$$

¹The ex ante distribution μ_0 describes the state-generating process. But each agent may have a different interpretation or additional pre-period information, which can allow heterogeneous starting assessments. By setting $\ell_{i,0} = \log \frac{\mu_0}{1-\mu_0}$ we can nest a common starting score as a special case.

²The communication system is treated as a fixed institutional or platform-level constraint rather than as an object of individual choice.

The decoded values are bounded. Write

$$\underline{r} = \min_{k \in \mathcal{M}} r_k, \quad \bar{r} = \max_{k \in \mathcal{M}} r_k.$$

We impose bin consistency:

$$r_1 \leq b_1, \quad r_k \in [b_{k-1}, b_k] \quad \text{for } k = 2, \dots, K-1, \quad r_K \geq b_{K-1}.$$

Bin consistency says that the decoded value assigned to a message lies in the score region represented by that message. It also implies that decoded values are weakly increasing:

$$r_1 \leq r_2 \leq \dots \leq r_K.$$

Because ψ is step-valued, within-bin movements in scores are socially invisible. A message changes only when a score crosses a cutpoint. We say that agent j has a cutpoint crossing between dates $t-1$ and t if

$$Q(\ell_{j,t}) \neq Q(\ell_{j,t-1}).$$

If no agent has a cutpoint crossing, then all messages and decoded values remain unchanged even if latent scores have moved.

Social exposure is represented by an $n \times n$ matrix $P = (P_{ij})_{i,j \in N}$. For each n , P is row-stochastic and has no self-weights:

$$P_{ij} \geq 0, \quad P_{ii} = 0, \quad \sum_{j=1}^n P_{ij} = 1 \quad \text{for every } i.$$

The coefficient P_{ij} is the weight that agent i assigns to agent j 's decoded value. The restriction $P_{ii} = 0$ reflects that each agent directly carries her own latent score in the updating rule below, while finite communication applies to information received from others.

At date t , agent i 's social aggregate is

$$S_{i,t} = \sum_{j=1}^n P_{ij} \psi(\ell_{j,t}).$$

Thus the network aggregates decoded values rather than latent scores. This distinction is central. Network mixing averages what agents communicate, even when their latent scores contain within-bin variation that is invisible to others.

Remark 1. Finite communication separates latent scores from social transmission. If two

score profiles ℓ and ℓ' satisfy $Q(\ell_i) = Q(\ell'_i)$ for every i , then the network observes the same messages, assigns the same decoded values, and forms the same social aggregates under both profiles even if $\ell_i \neq \ell'_i$. Thus within-bin score variation affects the agent who carries it, but it does not directly enter other agents' social inputs.

2.3 Private evidence and score dynamics

After date- t communication and before the next score is formed, the agent i receives private evidence $x_{i,t+1}$ with conditional densities $f_i(\cdot | \theta)$. We summarize the informational content of that evidence by a private increment $\Delta_{i,t+1}$ that can be written as the log-likelihood ratio

$$\Delta_{i,t+1} = \log \frac{f_i(x_{i,t+1} | \theta = 1)}{f_i(x_{i,t+1} | \theta = 0)}.$$

Positive increments favor $\theta = 1$, and negative increments favor $\theta = 0$.³

Assumption 1. *The private-increment process $\{\Delta_{i,t}\}_{i \in N, t \geq 1}$ satisfies the following conditions with constants $\bar{\Delta} < \infty$ and $\eta > 0$ that do not depend on n :*

- (i) *Conditional on θ , the family $\{\Delta_{i,t}\}_{i \in N, t \geq 1}$ is independent across agents and time.*
- (ii) *Private increments are bounded:*

$$|\Delta_{i,t}| \leq \bar{\Delta} \quad \text{a.s. for every } i, t.$$

- (iii) *Private increments are directionally informative:*

$$\mathbb{E}[\Delta_{i,t} | \theta = 1] \geq \eta, \quad \mathbb{E}[\Delta_{i,t} | \theta = 0] \leq -\eta \quad \text{for every } i, t.$$

Bounded private increments rule out one-shot revelation: no single private signal can move a score arbitrarily far in one period. Directional informativeness says that private increments have systematic drift toward the true state.

At the end of period t , agent i updates next period's score according to

$$\ell_{i,t+1} = (1 - \alpha)\ell_{i,t} + \alpha(S_{i,t} + \Delta_{i,t+1}), \quad \alpha \in (0, 1).$$

The parameter α measures responsiveness to current information. A smaller α gives more weight to the inherited score, while a larger α gives more weight to the current social and private inputs.

³The log-likelihood-ratio scale is used only to put private evidence on the same signed scale as the score.

In vector form,

$$\ell_{t+1} = (1 - \alpha)\ell_t + \alpha(P\psi(\ell_t) + \Delta_{t+1}),$$

where $\psi(\ell_t) = (\psi(\ell_{1,t}), \dots, \psi(\ell_{n,t}))^\top$ and $\Delta_{t+1} = (\Delta_{1,t+1}, \dots, \Delta_{n,t+1})^\top$.

The updating rule is implementable under finite communication. A received message reveals only that a neighbor's latent score lies in a communication bin. Exact Bayesian filtering would require beliefs over neighbors' unobserved score histories, private signals, and higher-order beliefs. The rule above instead uses the objects directly available in every period: the agent's inherited score, decoded values from others, and current private evidence.⁴

2.4 Timing and accuracy

Each period $t \geq 0$ has three stages. First, given current scores $(\ell_{j,t})_{j \in N}$, each agent j sends $m_{j,t} = Q(\ell_{j,t})$, receivers decode $y_{j,t} = \psi(\ell_{j,t})$, and agent i forms

$$S_{i,t} = \sum_{j=1}^n P_{ij} \psi(\ell_{j,t}).$$

Second, agent i observes private evidence summarized by $\Delta_{i,t+1}$. Third, agent i updates

$$\ell_{i,t+1} = (1 - \alpha)\ell_{i,t} + \alpha(S_{i,t} + \Delta_{i,t+1}).$$

Accuracy is measured on the probability scale. A score ℓ induces the probability assessment

$$\sigma(\ell) = \frac{1}{1 + \exp(-\ell)}.$$

The map σ is an evaluation map from scores to probability assessments. Agent i 's probability assessment at date t about θ is

$$a_{i,t} = \sigma(\ell_{i,t}).$$

Conditional on the realized state θ , define

$$\text{MSE}_t(\theta) = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[(a_{i,t} - \theta)^2 \mid \theta].$$

Equivalently, $-\text{MSE}_t(\theta)$ is the average payoff from reporting probability assessments under

⁴The updating rule can be viewed as an extension of the DeGroot model (DeGroot, 1974), where the social learning is given by $\ell_{t+1} = M\ell_t$, where M is a row-stochastic matrix. By setting $\Delta_{i,t} = 0, \forall i, t$ and $\psi(x) = x$, we can nest the DeGroot.

quadratic scoring loss. When $\theta = 1$, the term $(a_{i,t} - 1)^2$ vanishes as $\ell_{i,t} \rightarrow +\infty$. When $\theta = 0$, the term $a_{i,t}^2$ vanishes as $\ell_{i,t} \rightarrow -\infty$. We study long-run accuracy as $t \rightarrow \infty$ and compare accuracy across communication and network structures.

3 Continuous-communication benchmark

We first remove the finite-codebook friction while keeping the same private-increment process and the same score-updating rule. Under continuous communication, agents transmit their real-valued latent scores directly. Formally, replace the finite communication rule with the identity map, $\psi(\ell) = \ell$. The social aggregate vector is then $S_t = P\ell_t$, and the score vector follows

$$\ell_{t+1} = A\ell_t + \alpha\Delta_{t+1}, \quad A = (1 - \alpha)I + \alpha P,$$

where $\ell_t = (\ell_{1,t}, \dots, \ell_{n,t})^\top$ and $\Delta_t = (\Delta_{1,t}, \dots, \Delta_{n,t})^\top$.

For this benchmark, suppose that P is irreducible. Since P is row stochastic and irreducible, there is a unique stationary distribution $\pi \in \Delta(N)$ satisfying $\pi^\top P = \pi^\top$ by the Perron–Frobenius theorem.

Proposition 1. *Maintain Assumption 1, and suppose that P is irreducible. Under continuous communication, for every n and every agent $i \in N$,*

$$\ell_{i,t} \rightarrow +\infty \quad a.s. \text{ under } \theta = 1, \quad \ell_{i,t} \rightarrow -\infty \quad a.s. \text{ under } \theta = 0.$$

Consequently, $a_{i,t} \rightarrow \theta$ almost surely for every i , and $\text{MSE}_t(\theta) \rightarrow 0$.

Proposition 1 shows that the updating rule itself does not impose an information trap or ceiling. When scores are transmitted continuously, private increments enter future scores, those scores are transmitted to others, and the network repeatedly reallocates accumulated evidence across the society. In this sense, the benchmark is close to the DeGroot tradition of linear social learning, where social learning under a network can converge to the true state.⁵

This benchmark clarifies the source of the failure studied below. The obstacle is not the arrival of private evidence, nor the use of social averaging per se. It is the finite communication layer. Once agents transmit decoded messages rather than latent scores, a private increment affects social exposure only when it changes a finite message. The next section studies how this discreteness can turn fast mixing from an aggregating force into an absorbing force.

⁵See Golub and Jackson (2010). The benchmark differs from the canonical DeGroot model because agents continue to receive new private score increments over time.

Figure 1 illustrates the contrast between continuous score transmission and finite message transmission.

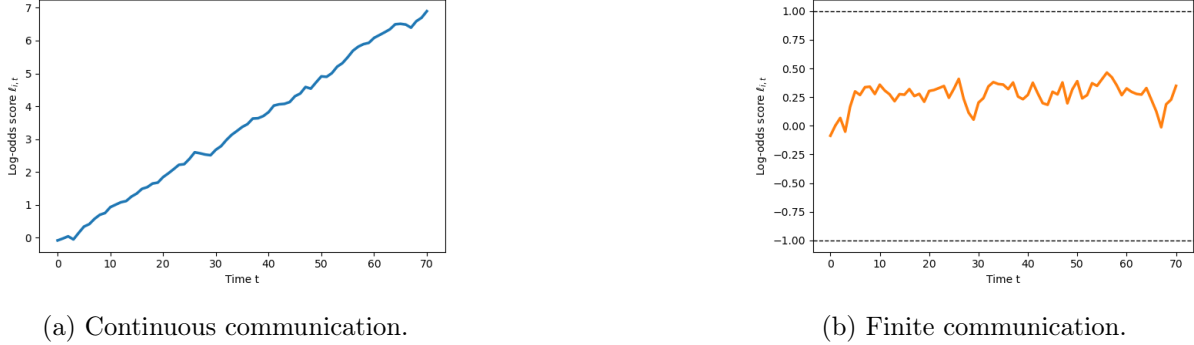


Figure 1: Learning process under continuous and finite communication when $\theta = 1$.

4 Fast Mixing and Same-Message Traps

Under continuous communication, social exposure averages latent scores. Under finite communication, it averages decoded messages. This distinction changes the role of mixing. Since the decoder is step-valued, movements of latent scores inside a bin are not socially transmitted. New communicated content appears only when some score crosses a cutpoint.

Fast mixing can therefore have a second effect besides information aggregation: it compresses cross-sectional disagreement before enough cutpoint crossings occur. Once latent scores are tightly clustered, a population mean that lies inside a communication bin can imply that almost all agents send the same message. If that same message set is locally stable, the common message becomes absorbing. This section formalizes the mechanism. The projected mixing coefficient $\lambda(P)$ measures the transmission margin: how much disagreement remains after one round of exposure.

4.1 Disagreement compression

For $v \in \mathbb{R}^n$, write

$$\bar{v} = \frac{1}{n} \sum_{i=1}^n v_i, \quad \Pi v = v - \bar{v} \mathbf{1}, \quad \|v\|_n = \left(\frac{1}{n} \sum_{i=1}^n v_i^2 \right)^{1/2},$$

where $\Pi = I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$. Thus Πv is the cross-sectional disagreement component of v . For the score profile $\ell_t = (\ell_{1,t}, \dots, \ell_{n,t})^\top$, define

$$D_t = \|\Pi \ell_t\|_n.$$

The relevant measure of mixing is the amount of disagreement that survives one round of exposure. Define

$$\lambda(P) = \sup_{\bar{v}=0, v \neq 0} \frac{\|\Pi P v\|_n}{\|v\|_n}.$$

Equivalently,

$$\|\Pi P v\|_n \leq \lambda(P) \|v\|_n \quad \text{for every } v \text{ with } \bar{v} = 0.$$

The coefficient has a direct exposure interpretation. Let $p_i^\top$ denote row i of P , and let

$$\bar{p}^\top = \frac{1}{n} \mathbf{1}^\top P$$

be the average exposure row. Since $\Pi P = P - \mathbf{1}\bar{p}^\top$, for every mean-zero disagreement vector v ,

$$(\Pi P v)_i = (p_i - \bar{p})^\top v.$$

Thus post-exposure disagreement is generated by differences across exposure rows. If exposure rows are similar, agents receive similar averages of any disagreement vector, so $\Pi P v$ is small. If exposure is segmented, $\lambda(P)$ is large.⁶

We say that P' is weakly faster mixing than P if

$$\lambda(P') \leq \lambda(P).$$

Thus, faster mixing means stronger one-step compression of the cross-sectional disagreement. Therefore, lower $\lambda(P)$ tightens the disagreement, $\|\Pi P v\|$ more.⁷

Let

$$\underline{r} = r_1, \quad \bar{r} = r_K, \quad R = \bar{r} - \underline{r}.$$

Define

$$\overline{D}(P) = \frac{\lambda(P)R}{2} + \bar{\Delta}$$

⁶Appendix B discusses the details.

⁷This is the notion of mixing relevant here because, in each period, decoded values are averaged once through P before scores are updated.

and

$$\hat{D}_t(P) = (1 - \alpha)^t D_0 + (1 - (1 - \alpha)^t) \bar{D}(P).$$

Lemma 1. *For every $t \geq 0$,*

$$D_t \leq \hat{D}_t(P).$$

Consequently, $\limsup_{t \rightarrow \infty} D_t \leq \bar{D}(P)$.

The lemma gives the first channel through which faster mixing matters. A smaller $\lambda(P)$ lowers $\bar{D}(P)$, and hence lowers the bound on cross-sectional score dispersion. Under finite communication, this is not only an aggregation force. It is also a compression force: it makes latent scores more similar before their within-bin movements become socially visible.

4.2 Buffered entry and local stability

Messages change only at cutpoints. Fix a bin

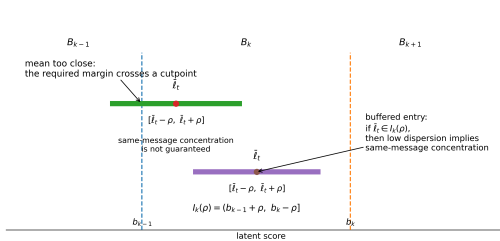
$$B_k = (b_{k-1}, b_k].$$

For $\rho > 0$, define the ρ -buffered interior of B_k by

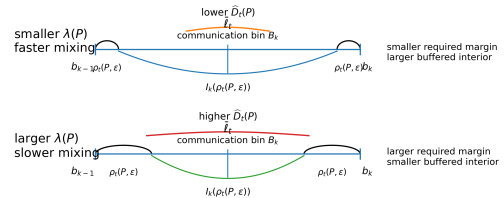
$$I_k(\rho) = (b_{k-1} + \rho, b_k - \rho],$$

with the convention that the lower buffer is ignored when $b_{k-1} = -\infty$, and the upper buffer is ignored when $b_k = +\infty$.

The buffer is the amount of room the mean needs in order for low dispersion to translate into common messages. If the mean lies close to a cutpoint, even a small amount of disagreement can place many agents on different sides of that cutpoint. If the mean lies inside a buffered interior, then low dispersion forces most agents into the same bin.



(a) Buffered mean for same-message concentration.



(b) Lower $\lambda(P)$ reduces the required buffer.

Figure 2: Disagreement compression, buffered interiors, and room for the mean.

For $\varepsilon \in (0, 1)$, define the required cutpoint buffer by

$$\rho_t(P, \varepsilon) = \frac{\widehat{D}_t(P)}{\sqrt{\varepsilon}}.$$

Let

$$C_{k,t} = \{i \in N : \ell_{i,t} \in B_k\}$$

be the set of agents who send message k at date t .

Lemma 2. *If $\bar{\ell}_t \in I_k(\rho_t(P, \varepsilon))$, then*

$$\frac{|C_{k,t}|}{n} \geq 1 - \varepsilon.$$

Lemma 2 identifies the threshold effect created by finite communication. Once cross-sectional disagreement is sufficiently compressed by Lemma 1, the message profile is largely determined by the position of the population mean relative to the cutpoints. If the mean lies inside a buffered part of bin B_k , then at most an ε -fraction of agents can lie outside that bin.

Thus $\rho_t(P, \varepsilon)$ is the distance from the adjacent cutpoints needed to turn the dispersion bound into same-message concentration. Since $\rho_t(P, \varepsilon)$ is increasing in $\widehat{D}_t(P)$, and $\widehat{D}_t(P)$ is increasing in $\lambda(P)$, faster mixing enlarges the set of mean scores that generate a large same-message group:

$$\lambda(P') \leq \lambda(P) \implies \rho_t(P', \varepsilon) \leq \rho_t(P, \varepsilon) \implies I_k(\rho_t(P, \varepsilon)) \subseteq I_k(\rho_t(P', \varepsilon)).$$

Hence, under faster mixing, the mean score need not lie as deeply inside a bin for most agents to send the same message. The reason is that Lemma 2 converts a dispersion bound into the same-message concentration.

Same-message concentration is not yet absorption. A large group may send the same message at date t , but its members can leave that bin at date $t + 1$ if they put enough weight on outsiders who send different messages. The relevant stability object is therefore outside exposure, not internal connectedness.

For $\xi \in [0, 1]$, define the worst-case input interval

$$\underline{h}_k(\xi) = (1 - \xi)r_k + \xi\underline{r} - \bar{\Delta}, \quad \bar{h}_k(\xi) = (1 - \xi)r_k + \xi\bar{r} + \bar{\Delta}.$$

This interval allows outside agents to send any decoded value and allows the private incre-

ment to take any value in $[-\bar{\Delta}, \bar{\Delta}]$.⁸

Definition 1. The bin $B_k = (b_{k-1}, b_k]$ is ξ -robust if

$$b_{k-1} < \underline{h}_k(\xi) \quad \text{whenever } b_{k-1} > -\infty,$$

and

$$\bar{h}_k(\xi) \leq b_k \quad \text{whenever } b_k < +\infty.$$

Thus a ξ -robust bin remains stable for a same-message group whose outside exposure is at most ξ . If an agent starts in B_k , receives current input in B_k , and updates by a convex combination of the two, then her next score remains in B_k .

4.3 Same-message traps and accuracy loss

The previous subsection showed how faster mixing affects same-message concentration. We now consider the stability needed to convert same-message concentration into absorption.

The stability condition is stated as a bounded row-diffuseness restriction. Define

$$\omega(P) = n \max_{i,j \in N} P_{ij}.$$

For $\bar{\omega} < \infty$, let

$$\mathcal{P}_n(\bar{\omega}) = \{P : \omega(P) \leq \bar{\omega}\}.$$

This class rules out exposure matrices in which a small set of agents can receive disproportionately large one-period weight. The restriction is in the spirit of the no-prominent-agent conditions used in large-society DeGroot learning, where successful aggregation requires that no individual retain macroscopic influence as the society grows (Golub and Jackson, 2010). Here the condition is used for a different purpose: not to guarantee wisdom of crowds, but to ensure that once an ε -large same-message group has formed, the remaining outsiders cannot destabilize it through concentrated exposure.⁹

⁸This is a worst-case interval because it does not use the realized outside messages or the realized private increment. It only uses the bounds that are known uniformly. If a member of the same-message group places at most ξ weight outside the group, then at least $1 - \xi$ of her social input comes from agents sending message k , contributing r_k . The remaining outside weight ξ could be assigned to the lowest decoded value $\underline{r}$ or to the highest decoded value $\bar{r}$. The private increment could also take any value in $[-\bar{\Delta}, \bar{\Delta}]$. Therefore every possible current input for such an agent lies in $[\underline{h}_k(\xi), \bar{h}_k(\xi)]$. The interval is worst-case in the sense that it covers all admissible outside message profiles and all admissible private increments, not just the realized continuation.

⁹If $P \in \mathcal{P}_n(\bar{\omega})$ and $C \subseteq N$ satisfies $|C|/n \geq 1 - \varepsilon$, then $\max_{i \in C} \sum_{j \notin C} P_{ij} \leq \bar{\omega}\varepsilon$. Thus any ε -large same-message group has outside exposure at most $\bar{\xi}(\bar{\omega}, \varepsilon) = \min\{\bar{\omega}\varepsilon, 1\}$.

Theorem 1. *Maintain Assumption 1. Fix a date T , a bin B_k , and $\varepsilon \in (0, 1)$. Suppose $P \in \mathcal{P}_n(\bar{\omega})$. Let $\rho_T(P, \varepsilon) = \frac{\hat{D}_T(P)}{\sqrt{\varepsilon}}$, and $\bar{\xi}(\bar{\omega}, \varepsilon) = \min\{\bar{\omega}\varepsilon, 1\}$. If $\bar{\ell}_T \in I_k(\rho_T(P, \varepsilon))$ and B_k is $\bar{\xi}(\bar{\omega}, \varepsilon)$ -robust, then the set*

$$C_T = \{i \in N : \ell_{i,T} \in B_k\}$$

satisfies

$$\frac{|C_T|}{n} \geq 1 - \varepsilon,$$

and every agent in C_T remains in bin B_k forever:

$$\ell_{i,t} \in B_k \quad \text{and} \quad Q(\ell_{i,t}) = k$$

for every $i \in C_T$ and every $t \geq T$.

The theorem identifies how fast mixing can produce a learning trap under finite communication. The point is not that mixing is always harmful. The point is that lower $\lambda(P)$ compresses scores and lowers the cutpoint margin needed for a common message. If the compressed population enters a robust but insufficiently informative bin, the finite codebook prevents within-bin private learning from being socially transmitted, and the common message persists.

Corollary 1. *Fix T , k , ε , and a row-diffuseness bound $\bar{\omega} < \infty$. Let $\bar{\xi}(\bar{\omega}, \varepsilon) = \min\{\bar{\omega}\varepsilon, 1\}$. Suppose P' and P are two exposure matrices on the same agent set with*

$$\omega(P') \leq \bar{\omega}, \quad \omega(P) \leq \bar{\omega}, \quad \lambda(P') \leq \lambda(P).$$

Then $\rho_T(P', \varepsilon) \leq \rho_T(P, \varepsilon)$ and $I_k(\rho_T(P, \varepsilon)) \subseteq I_k(\rho_T(P', \varepsilon))$.

Corollary 1 is the formal sense in which faster mixing makes trap entry easier. A lower $\lambda(P)$ reduces the cutpoint buffer needed for the population mean to generate same-message concentration. The corollary does not say that the realized trajectory under P' enters the same bin as the trajectory under P . That is the selection problem studied in the next section. It says that, conditional on candidate entry into a bin, faster mixing expands the mean-score region from which finite communication can turn same-message concentration into absorption.

Same-message absorption affects accuracy when the absorbing bin is not sufficiently informative about the state. For each bin $B_k = (b_{k-1}, b_k]$, define

$$e_k(1) = \begin{cases} (1 - \sigma(b_k))^2, & b_k < +\infty, \\ 0, & b_k = +\infty, \end{cases} \quad e_k(0) = \begin{cases} \sigma(b_{k-1})^2, & b_{k-1} > -\infty, \\ 0, & b_{k-1} = -\infty. \end{cases}$$

Thus $e_k(\theta)$ is the smallest squared error that can be guaranteed when an agent's score remains in bin B_k . If B_k is an interior bin,¹⁰ meaning that both cutpoints are finite, then

$$e_k(1) > 0 \quad \text{and} \quad e_k(0) > 0.$$

Corollary 2. *Suppose $C \subseteq N$ satisfies $|C|/n \geq 1 - \varepsilon$. For a date T and a bin B_k , define the trapping event $E_{C,T,k} = \{\ell_{i,t} \in B_k \text{ for every } i \in C \text{ and every } t \geq T\}$. Then, for every $t \geq T$ and every $\theta \in \{0, 1\}$,*

$$\text{MSE}_t(\theta) \geq \Pr(E_{C,T,k} \mid \theta)(1 - \varepsilon)e_k(\theta).$$

If B_k is an interior bin, then $e_k(0) > 0$ and $e_k(1) > 0$. Hence a large trapped core that persists with positive conditional probability generates a strictly positive error lower bound in the corresponding state, and a large trapped core that persists almost surely generates a strictly positive long-run error lower bound in both states.

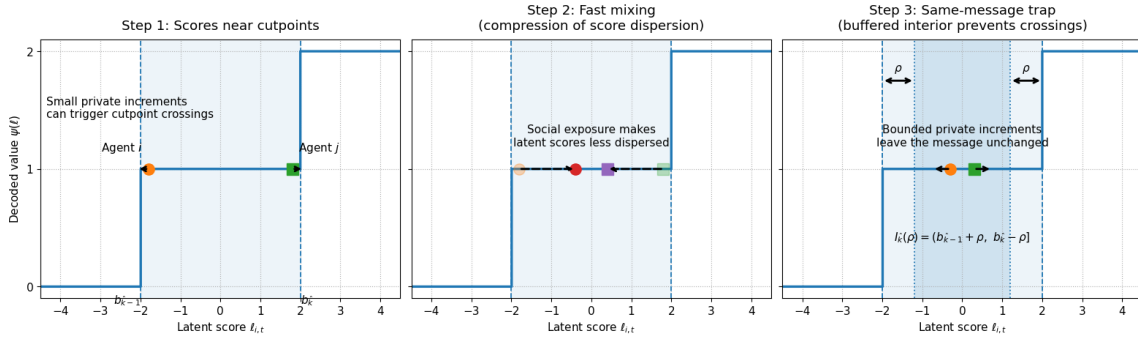


Figure 3: Fast mixing mechanism under finite codebook.

The general trap theorem explains when a same-message region is absorbing, but it does not determine which region is reached. Selection is harder because finite communication makes the dynamics path dependent: the current message profile determines social exposure, social exposure determines the next set of threshold crossings, and those crossings determine future exposure. We therefore turn to a block environment in which this selection problem can be solved explicitly.

5 Block Mixing and Endogenous Message Selection

The previous section gives the general absorption logic. Once finite communication has compressed learning into message regions, a sufficiently large and cohesive group can remain

¹⁰When $K \geq 3$, the interior bins are $k = 2, \dots, K - 1$.

in a robust bin forever.

This section studies message selection in a canonical block environment.¹¹ The block structure creates a simple tradeoff. One group initially sends the state-aligned message, while another group is lagging. More cross-group exposure helps the lagging group move toward the state-aligned message. But the same exposure also pulls the initially state-aligned group toward the lagging group.

Throughout this section, fix $\theta = 1$. The case $\theta = 0$ is symmetric after reversing signs and exchanging low and high messages.

5.1 The canonical block environment

There are three messages. Fix $c > 0$ and define

$$B_1 = (-\infty, -c], \quad B_2 = (-c, c], \quad B_3 = (c, \infty).$$

The decoded values are symmetric:

$$r_1 = -r, \quad r_2 = 0, \quad r_3 = r,$$

where $r > 0$. Under $\theta = 1$, message 3 is state-aligned, message 1 is state-opposed, and message 2 is interior.¹²

Assume

$$\bar{\Delta} < c < r - \bar{\Delta}.$$

The first inequality makes interior consensus robust: if all agents send message 2, then every current social-plus-private input lies in $(-c, c)$. The second inequality makes high consensus robust: if all agents send message 3, then even the lowest possible current input remains above c . By symmetry, low consensus is also robust.

Fix a finite society N , and partition it into two nonempty groups H and L . Let $|H| \geq 2$,

¹¹Block networks are a standard way to represent homophily, modularity, and group-based exposure. See, for example, [Holland et al. \(1983\)](#) on stochastic block models and [Currarini et al. \(2009\)](#) and [Golub and Jackson \(2012\)](#) on homophily and social learning in economic networks. We use the block structure here as a reduced-form device for varying cross-group exposure while preserving a transparent two-group geometry.

¹²The three-message codebook is the minimal codebook that displays the phase diagram transparently. Richer threshold codebooks do not remove the mechanism. If a K -message codebook contains more than three messages that play the roles of state-opposed, interior, and state-aligned messages, and if the corresponding containment inequalities below hold for those bins and decoded values, the same pathwise conclusions follow along the specified path.

$|L| \geq 2$, and

$$q = \frac{|H|}{n} \in \left(\frac{1}{2}, 1\right).$$

The blocks determine exposure, not the direct observability of latent scores: agents observe block membership and communicated messages, but they do not observe other agents' exact scores. We impose block-constant initial scores,

$$\ell_{i,0} = h_0 > c \quad \text{for } i \in H, \quad \ell_{i,0} = g_0 < -c \quad \text{for } i \in L.$$

Thus group H is initially state-aligned and sends message 3, while group L is initially lagging and sends message 1. The initial block message profile is therefore $(3, 1)$.¹³ Initial message heterogeneity is essential. If all agents initially sent the same robust message, then everyone would receive the same decoded social input regardless of the mixing parameter, and mixing would have no selection role.

The parameter $\zeta \in [0, 1]$ measures cross-group mixing. The exposure matrix P^ζ has no self-weights and is defined by

$$P_{ij}^\zeta = \begin{cases} \frac{1 - \zeta(1 - q)}{|H| - 1}, & i, j \in H, i \neq j, \\ \frac{\zeta(1 - q)}{|L|}, & i \in H, j \in L, \\ \frac{\zeta q}{|H|}, & i \in L, j \in H, \\ \frac{1 - \zeta q}{|L| - 1}, & i, j \in L, i \neq j, \\ 0, & i = j. \end{cases}$$

An agent in H assigns total weight $\zeta(1 - q)$ to group L , while an agent in L assigns total weight ζq to group H . Since $q > 1/2$, the initially state-aligned group is the larger block.

For block-level calculations, the relevant exposure matrix is

$$\mathcal{P}^\zeta = \begin{pmatrix} 1 - \zeta(1 - q) & \zeta(1 - q) \\ \zeta q & 1 - \zeta q \end{pmatrix}.$$

¹³Appendix C.2 considers the initial profile $(3, 2)$ as a robustness check.

When $\zeta = 0$, there is no cross-group exposure. When $\zeta = 1$,

$$\mathcal{P}^1 = \begin{pmatrix} q & 1-q \\ q & 1-q \end{pmatrix},$$

so both groups face the same aggregate mixture of H - and L -messages. Thus larger ζ means stronger cross-group averaging and faster compression of group differences.¹⁴

If all agents in H send decoded value y_H and all agents in L send decoded value y_L , then the block social aggregates are

$$S_H(\zeta; y_H, y_L) = (1 - \zeta(1 - q))y_H + \zeta(1 - q)y_L, \quad S_L(\zeta; y_H, y_L) = \zeta q y_H + (1 - \zeta q)y_L.$$

Starting from the initial message profile $(3, 1)$, the decoded values are $y_H = r$ and $y_L = -r$. Hence

$$s_H^{31}(\zeta) = r(1 - 2(1 - q)\zeta), \quad s_L^{31}(\zeta) = r(2q\zeta - 1).$$

The first expression is decreasing in ζ , while the second is increasing in ζ . This is the basic selection tradeoff. More mixing raises the lagging group's exposure to the state-aligned message, but it also lowers the initially state-aligned group's exposure to its own message.

5.2 The phase diagram

We now define three robust regions of the mixing parameter. Let

$$s_H^{31}(\zeta) = r(1 - 2\zeta(1 - q)), \quad s_L^{31}(\zeta) = r(2\zeta q - 1),$$

and

$$s_H^{32}(\zeta) = r(1 - \zeta(1 - q)), \quad s_L^{32}(\zeta) = \zeta q r.$$

Define

$$h_1^\pm(\zeta) = (1 - \alpha)h_0 + \alpha(s_H^{31}(\zeta) \pm \bar{\Delta}), \quad g_1^\pm(\zeta) = (1 - \alpha)g_0 + \alpha(s_L^{31}(\zeta) \pm \bar{\Delta}),$$

and

$$h_2^-(\zeta) = (1 - \alpha)h_1^-(\zeta) + \alpha(s_H^{32}(\zeta) - \bar{\Delta}), \quad g_2^-(\zeta) = (1 - \alpha)g_1^-(\zeta) + \alpha(s_L^{32}(\zeta) - \bar{\Delta}).$$

¹⁴Appendix C.1 discusses the details of the faster mixing.

The robust segmentation, moderate-mixing, and interior-consensus regions are

$$\mathcal{Z}_S = \{\zeta \in [0, 1] : s_H^{31}(\zeta) - \bar{\Delta} > c, s_L^{31}(\zeta) + \bar{\Delta} \leq -c\},$$

$$\mathcal{Z}_M = \{\zeta \in [0, 1] : h_1^-(\zeta) > c, -c < g_1^-(\zeta) \leq g_1^+(\zeta) \leq c, h_2^-(\zeta) > c, g_2^-(\zeta) > c\},$$

and

$$\mathcal{Z}_I = \{\zeta \in [0, 1] : -c < h_1^-(\zeta) \leq h_1^+(\zeta) \leq c, -c < g_1^-(\zeta) \leq g_1^+(\zeta) \leq c\}.$$

The first preserves segmentation, the second selects high-message consensus, and the third selects interior-message consensus.

Theorem 2. *Consider the block environment above, and maintain $\bar{\Delta} < c < r - \bar{\Delta}$. Along every realization satisfying $|\Delta_{i,t}| \leq \bar{\Delta}$ for every $i \in N$ and $t \geq 1$, the following statements hold.*

1. *If $\zeta \in \mathcal{Z}_S$, then the initial profile $(3, 1)$ is absorbing. Group H sends message 3 forever, and group L sends message 1 forever.*

2. *If $\zeta \in \mathcal{Z}_M$, then the path*

$$(3, 1) \rightarrow (3, 2) \rightarrow (3, 3)$$

is selected. All agents send message 3 from date 2 onward.

3. *If $\zeta \in \mathcal{Z}_I$, then the path*

$$(3, 1) \rightarrow (2, 2)$$

is selected. All agents send message 2 from date 1 onward.

4. *The regions are ordered by the mixing parameter. Whenever $\zeta_S \in \mathcal{Z}_S$, $\zeta_M \in \mathcal{Z}_M$, and $\zeta_I \in \mathcal{Z}_I$, we have*

$$\zeta_S < \zeta_M < \zeta_I.$$

In the robust regions identified by the theorem, low mixing preserves the initial segmentation. Moderate mixing gives group L enough exposure to move upward while keeping group H in the high bin, so the society reaches high-message consensus. Excessive mixing averages the two groups too quickly and pulls group H into the interior before group L reaches the high bin.

The robust phase regions have an immediate accuracy implication. Write $\text{MSE}_t(1; \zeta)$ for mean squared error under P^ζ , conditional on $\theta = 1$. Under this state, the high bin is state-aligned, whereas the interior bin is less informative. Hence moderate mixing can select a more accurate absorbing message than excessive mixing.

Corollary 3. *If $\zeta_M \in \mathcal{Z}_M$ and $\zeta_I \in \mathcal{Z}_I$, then, conditional on $\theta = 1$,*

$$\text{MSE}_t(1; \zeta_M) < \text{MSE}_t(1; \zeta_I) \quad \text{for every } t \geq 2.$$

Thus more mixing need not improve accuracy. Moderate mixing helps because the initially state-aligned group pulls the lagging group across the relevant cutpoints while remaining in the high bin. Excessive mixing hurts because cross-group averaging becomes strong enough to select the interior bin before high-message consensus is reached.

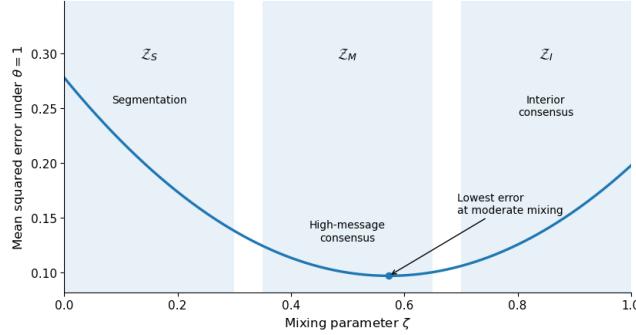


Figure 4: Under $\theta = 1$, moderate mixing selects high-message consensus and minimizes error. Low mixing preserves segmented communication, while excessive mixing selects interior consensus.

5.3 Timing of integration

The phase diagram gives a timing interpretation of integration. High integration is harmful when it arrives before the state-aligned message has become socially visible. It is no longer harmful after high-message consensus has formed. To make this point explicit, allow the mixing parameter to vary deterministically over time. Given a schedule

$$\zeta = (\zeta_0, \zeta_1, \zeta_2, \dots),$$

let the date- t exposure matrix be P^{ζ_t} , so that

$$\ell_{i,t+1} = (1 - \alpha)\ell_{i,t} + \alpha \left(\sum_{j=1}^n P_{ij}^{\zeta_t} \psi(\ell_{j,t}) + \Delta_{i,t+1} \right).$$

Write $\text{MSE}_t(1; \zeta)$ for mean squared error under the schedule ζ , conditional on $\theta = 1$.

Remark 2. Suppose that $\zeta_M \in \mathcal{Z}_M$ and $\zeta_I \in \mathcal{Z}_I$. Consider the two schedules

$$\zeta^{\text{imm}} = (\zeta_I, \zeta_I, \zeta_I, \dots), \quad \zeta^{\text{del}} = (\zeta_M, \zeta_M, \zeta_I, \zeta_I, \dots).$$

Then, along every realization satisfying the private-increment bound, the following statements hold.

1. Under ζ^{imm} , the path $(3, 1) \rightarrow (2, 2)$ is selected, and all agents send message 2 from date 1 onward.
2. Under ζ^{del} , the path $(3, 1) \rightarrow (3, 2) \rightarrow (3, 3)$ is selected, and all agents send message 3 from date 2 onward.
3. Conditional on $\theta = 1$,

$$\text{MSE}_t(1; \zeta^{\text{del}}) < \text{MSE}_t(1; \zeta^{\text{imm}}) \quad \text{for every } t \geq 2.$$

The result gives the timing implication of the phase diagram. Immediate high integration can be harmful because it compresses the two groups into the interior message before the high message has spread. Moderate early integration allows the low-message group to move upward while the high-message group remains in the state-aligned bin. Once high-message consensus has formed, further integration is no longer harmful. If all agents send message 3, then every row-stochastic exposure matrix generates the same social aggregate r , so the high bin remains absorbing.

6 Conclusion

This paper studies how finite communication changes the value of network mixing. Agents carry real-valued latent assessment scores, but the network observes only finite messages generated by a common codebook. This distinction matters. When communication is continuous, the same score-updating rule aggregates information and every agent learns the true binary state. When communication is finite, social exposure depends only on decoded messages, so within-bin score movements remain socially invisible. Information affects the social environment only when it moves scores across message thresholds.

The main mechanism is that mixing can become an absorbing force rather than only an aggregating force. Faster mixing compresses cross-sectional disagreement, but under finite communication this compression can place many agents in the same message region before the state-aligned message has become socially visible. We characterize when this disagreement compression becomes a same-message trap and show how absorption creates persistent accuracy loss. In the block environment, the same mechanism selects among messages: low mixing preserves segmentation, moderate mixing selects state-aligned consensus, and excessive mixing prematurely selects an interior message. Accuracy is therefore nonmonotone in

the amount of cross-group exposure.

The results suggest that the value of integration depends on timing. More exposure is beneficial after informative assessments have crossed public thresholds, but it can be harmful before those assessments are visible through the communication system. Thus, in environments organized around ratings, grades, risk tiers, classifications, or platform labels, the design problem is not simply whether to increase mixing. It is when to increase mixing, given that finite messages reveal only threshold crossings and hide the latent movements that occur within categories.

A Proofs

A.1 Proofs of main-text results

Proof of Proposition 1

Under continuous communication, the score vector satisfies

$$\ell_t = A^t \ell_0 + \alpha \sum_{s=1}^t A^{t-s} \Delta_s, \quad A = (1 - \alpha)I + \alpha P.$$

Since P is irreducible and $0 < \alpha < 1$, the matrix A is irreducible and has a positive diagonal. Hence A is primitive. Let π be the unique stationary distribution of P . Then $\pi^\top A = \pi^\top$, and

$$A^k \rightarrow \mathbf{1}\pi^\top.$$

Thus there exist constants $C < \infty$ and $\rho \in (0, 1)$ such that

$$\|A^k - \mathbf{1}\pi^\top\| \leq C\rho^k \quad \text{for every } k \geq 0.$$

Fix $\theta = 1$. Let $Z_s = \pi^\top \Delta_s$. By Assumption 1, the variables $\{Z_s\}_{s \geq 1}$ are conditionally independent under $\theta = 1$ and uniformly bounded. Moreover,

$$\mathbb{E}[Z_s \mid \theta = 1] = \sum_{i=1}^n \pi_i \mathbb{E}[\Delta_{i,s} \mid \theta = 1] \geq \eta \sum_{i=1}^n \pi_i = \eta.$$

The strong law for independent uniformly bounded variables gives

$$\frac{1}{t} \sum_{s=1}^t (Z_s - \mathbb{E}[Z_s \mid \theta = 1]) \rightarrow 0 \quad \text{a.s. under } \theta = 1.$$

Hence

$$\sum_{s=1}^t \pi^\top \Delta_s = \sum_{s=1}^t Z_s \geq \eta t + o(t) \rightarrow +\infty \quad \text{a.s. under } \theta = 1.$$

It remains to verify that replacing A^{t-s} by $\mathbf{1}\pi^\top$ changes the accumulated evidence by only a bounded term. Since private increments are uniformly bounded and n is fixed, there is $M < \infty$ such that $\|\Delta_s\| \leq M$ for every s . Therefore

$$\left\| \sum_{s=1}^t (A^{t-s} - \mathbf{1}\pi^\top) \Delta_s \right\| \leq \sum_{s=1}^t C \rho^{t-s} M \leq \frac{CM}{1-\rho}.$$

Thus

$$\sum_{s=1}^t A^{t-s} \Delta_s = \mathbf{1} \sum_{s=1}^t \pi^\top \Delta_s + O(1),$$

where the $O(1)$ term is uniform in t . Since $A^t \ell_0 \rightarrow \mathbf{1}\pi^\top \ell_0$, the initial-condition term is bounded. Because $\alpha > 0$, each coordinate of ℓ_t diverges to $+\infty$ almost surely under $\theta = 1$.

The proof under $\theta = 0$ is symmetric. In that state,

$$\mathbb{E}[\pi^\top \Delta_s \mid \theta = 0] = \sum_{i=1}^n \pi_i \mathbb{E}[\Delta_{i,s} \mid \theta = 0] \leq -\eta.$$

The same strong-law argument gives

$$\sum_{s=1}^t \pi^\top \Delta_s \rightarrow -\infty \quad \text{a.s. under } \theta = 0.$$

The same geometric bound then implies $\ell_{i,t} \rightarrow -\infty$ almost surely for every i . Since $\sigma(\ell) \rightarrow 1$ as $\ell \rightarrow +\infty$ and $\sigma(\ell) \rightarrow 0$ as $\ell \rightarrow -\infty$, we have $a_{i,t} \rightarrow \theta$ almost surely for every i . Bounded convergence then implies $\text{MSE}_t(\theta) \rightarrow 0$. ■

Proof of Lemma 1

Let $y_t = (\psi(\ell_{1,t}), \dots, \psi(\ell_{n,t}))^\top$. The score recursion is

$$\ell_{t+1} = (1 - \alpha)\ell_t + \alpha P y_t + \alpha \Delta_{t+1}.$$

Applying Π and using the triangle inequality gives

$$D_{t+1} \leq (1 - \alpha)D_t + \alpha \|\Pi P y_t\|_n + \alpha \|\Pi \Delta_{t+1}\|_n.$$

Because $P\mathbf{1} = \mathbf{1}$, we have $\Pi Py_t = \Pi P \Pi y_t$. By the definition of $\lambda(P)$,

$$\|\Pi Py_t\|_n \leq \lambda(P) \|\Pi y_t\|_n.$$

Since decoded values lie in $[\underline{r}, \bar{r}]$, Lemma 3 gives $\|\Pi y_t\|_n \leq R/2$. Since $|\Delta_{i,t+1}| \leq \bar{\Delta}$, the same range bound gives $\|\Pi \Delta_{t+1}\|_n \leq \bar{\Delta}$. Therefore

$$D_{t+1} \leq (1 - \alpha)D_t + \alpha \left(\frac{\lambda(P)R}{2} + \bar{\Delta} \right) = (1 - \alpha)D_t + \alpha \bar{D}(P).$$

Iterating this recursion yields

$$D_t \leq (1 - \alpha)^t D_0 + (1 - (1 - \alpha)^t) \bar{D}(P) = \hat{D}_t(P).$$

Since $\hat{D}_t(P) \rightarrow \bar{D}(P)$, it follows that

$$\limsup_{t \rightarrow \infty} D_t \leq \bar{D}(P).$$

■

Proof of Lemma 2

By Lemma 1, $D_t \leq \hat{D}_t(P)$. Write $\rho = \rho_t(P, \varepsilon) = \hat{D}_t(P)/\sqrt{\varepsilon}$, and let

$$A_t = \{i \in N : \ell_{i,t} \notin B_k\}.$$

If $\hat{D}_t(P) = 0$, then $D_t = 0$, so all scores equal $\bar{\ell}_t$. Since $\bar{\ell}_t \in I_k(0)$, all agents lie in B_k , and the conclusion follows.

Suppose next that $\hat{D}_t(P) > 0$. If $i \in A_t$, then $\ell_{i,t}$ lies outside B_k . Since $\bar{\ell}_t \in I_k(\rho)$, and using the one-sided buffer conventions for the two extreme bins, every such i satisfies

$$|\ell_{i,t} - \bar{\ell}_t| \geq \rho.$$

Therefore

$$D_t^2 = \frac{1}{n} \sum_{i=1}^n (\ell_{i,t} - \bar{\ell}_t)^2 \geq \frac{|A_t|}{n} \rho^2.$$

Together with $D_t \leq \hat{D}_t(P)$ and $\rho^2 = \hat{D}_t(P)^2/\varepsilon$, this gives

$$\frac{|A_t|}{n} \leq \varepsilon.$$

Hence at least a fraction $1 - \varepsilon$ of agents lie in B_k , that is,

$$\frac{|C_{k,t}|}{n} \geq 1 - \varepsilon.$$

■

Proof of Theorem 1

By Lemma 2, the set

$$C_T = \{i \in N : \ell_{i,T} \in B_k\}$$

satisfies

$$\frac{|C_T|}{n} \geq 1 - \varepsilon.$$

By Lemma 4,

$$\sum_{j \notin C_T} P_{ij} \leq \omega(P)\varepsilon \quad \text{for every } i \in N.$$

Since $P \in \mathcal{P}_n(\bar{\omega})$, we have $\omega(P) \leq \bar{\omega}$. Total outside exposure is also at most one, so for every $i \in C_T$,

$$\sum_{j \notin C_T} P_{ij} \leq \min\{\bar{\omega}\varepsilon, 1\} = \bar{\xi}(\bar{\omega}, \varepsilon).$$

Because B_k is $\bar{\xi}(\bar{\omega}, \varepsilon)$ -robust, Proposition 2 implies that every agent in C_T remains in B_k and sends message k at every date $t \geq T$. ■

Proof of Corollary 1

Since $\lambda(P') \leq \lambda(P)$,

$$\bar{D}(P') = \frac{\lambda(P')R}{2} + \bar{\Delta} \leq \frac{\lambda(P)R}{2} + \bar{\Delta} = \bar{D}(P).$$

The two exposure matrices are compared on the same agent set and with the same initial score profile, so D_0 is common. Hence

$$\hat{D}_T(P') = (1 - \alpha)^T D_0 + (1 - (1 - \alpha)^T) \bar{D}(P') \leq \hat{D}_T(P).$$

Dividing by $\sqrt{\varepsilon}$ gives

$$\rho_T(P', \varepsilon) \leq \rho_T(P, \varepsilon).$$

A smaller buffer enlarges the buffered interior. Therefore, with the same one-sided conventions for the extreme bins,

$$I_k(\rho_T(P, \varepsilon)) \subseteq I_k(\rho_T(P', \varepsilon)).$$

■

Proof of Corollary 2

Fix a state $\theta \in \{0, 1\}$. Let

$$E_{C,T,k} = \{\ell_{i,s} \in B_k \text{ for every } i \in C \text{ and every } s \geq T\}$$

be the trapping event. We first prove the pointwise loss bound on this event.

Take any realization $\omega \in E_{C,T,k}$ and any $i \in C$. Then $\ell_{i,t}(\omega) \in B_k$ for every $t \geq T$.

Suppose first that $\theta = 1$. If $b_k < +\infty$, then $\ell_{i,t}(\omega) \leq b_k$, so by monotonicity of σ ,

$$a_{i,t}(\omega) = \sigma(\ell_{i,t}(\omega)) \leq \sigma(b_k).$$

Hence

$$(a_{i,t}(\omega) - 1)^2 = (1 - a_{i,t}(\omega))^2 \geq (1 - \sigma(b_k))^2 = e_k(1).$$

If $b_k = +\infty$, then $e_k(1) = 0$, so the same inequality is trivial.

Now suppose that $\theta = 0$. If $b_{k-1} > -\infty$, then $\ell_{i,t}(\omega) > b_{k-1}$, and therefore

$$a_{i,t}(\omega) = \sigma(\ell_{i,t}(\omega)) > \sigma(b_{k-1}).$$

Thus

$$(a_{i,t}(\omega) - 0)^2 = a_{i,t}(\omega)^2 \geq \sigma(b_{k-1})^2 = e_k(0).$$

If $b_{k-1} = -\infty$, then $e_k(0) = 0$, and the inequality is again trivial.

Therefore, for every $\theta \in \{0, 1\}$, every $t \geq T$, every $\omega \in E_{C,T,k}$, and every $i \in C$,

$$(a_{i,t}(\omega) - \theta)^2 \geq e_k(\theta).$$

Consequently, on $E_{C,T,k}$,

$$\frac{1}{n} \sum_{i=1}^n (a_{i,t}(\omega) - \theta)^2 \geq \frac{1}{n} \sum_{i \in C} (a_{i,t}(\omega) - \theta)^2 \geq \frac{|C|}{n} e_k(\theta) \geq (1 - \varepsilon) e_k(\theta).$$

Equivalently, for every realization,

$$\frac{1}{n} \sum_{i=1}^n (a_{i,t} - \theta)^2 \geq (1 - \varepsilon) e_k(\theta) \mathbf{1}_{E_{C,T,k}}.$$

Taking expectations conditional on θ gives

$$\text{MSE}_t(\theta) = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[(a_{i,t} - \theta)^2 \mid \theta] \geq \Pr(E_{C,T,k} \mid \theta) (1 - \varepsilon) e_k(\theta).$$

In particular, if the trapping property holds almost surely conditional on θ , that is, if $\Pr(E_{C,T,k} \mid \theta) = 1$, then

$$\text{MSE}_t(\theta) \geq (1 - \varepsilon) e_k(\theta) \quad \text{for every } t \geq T.$$

If B_k is an interior bin, then both cutpoints are finite, so $e_k(1) > 0$ and $e_k(0) > 0$. Hence an almost-sure large trapped core implies a strictly positive long-run error lower bound in both states. ■

Proof of Theorem 2

We first prove the three pathwise selection statements. Fix $\zeta \in \mathcal{Z}_S$. At date 0, the block message profile is $(3, 1)$. Suppose that the profile is $(3, 1)$ at some date t . Then agents in H receive social aggregate $s_H^{31}(\zeta)$, while agents in L receive social aggregate $s_L^{31}(\zeta)$. Since $\zeta \in \mathcal{Z}_S$,

$$s_H^{31}(\zeta) - \bar{\Delta} > c, \quad s_L^{31}(\zeta) + \bar{\Delta} \leq -c.$$

Thus, for every admissible private increment, $s_H^{31}(\zeta) + \Delta_{i,t+1} > c$ for $i \in H$, and $s_L^{31}(\zeta) + \Delta_{i,t+1} \leq -c$ for $i \in L$. If $i \in H$ and $\ell_{i,t} > c$, then

$$\ell_{i,t+1} = (1 - \alpha)\ell_{i,t} + \alpha(s_H^{31}(\zeta) + \Delta_{i,t+1}) > c,$$

because the update is a convex combination of two numbers strictly above c . If $i \in L$ and $\ell_{i,t} \leq -c$, then

$$\ell_{i,t+1} = (1 - \alpha)\ell_{i,t} + \alpha(s_L^{31}(\zeta) + \Delta_{i,t+1}) \leq -c,$$

because the update is a convex combination of two numbers weakly below $-c$. Hence the profile $(3, 1)$ reproduces itself. Since the initial profile is $(3, 1)$, induction implies that group H sends message 3 forever and group L sends message 1 forever.

Next fix $\zeta \in \mathcal{Z}_M$. Starting from $(3, 1)$, every H -agent satisfies

$$\ell_{i,1} \geq (1 - \alpha)h_0 + \alpha(s_H^{31}(\zeta) - \bar{\Delta}) = h_1^-(\zeta) > c,$$

so every H -agent sends message 3 at date 1. Every L -agent satisfies

$$-c < g_1^-(\zeta) \leq \ell_{i,1} \leq g_1^+(\zeta) \leq c,$$

so every L -agent sends message 2 at date 1. Hence the date-one block message profile is $(3, 2)$. Given this profile, the block social aggregates are $s_H^{32}(\zeta)$ and $s_L^{32}(\zeta)$. For $i \in H$ and $i \in L$, respectively,

$$\ell_{i,2} \geq (1 - \alpha)h_1^-(\zeta) + \alpha(s_H^{32}(\zeta) - \bar{\Delta}) = h_2^-(\zeta) > c, \quad \ell_{i,2} \geq (1 - \alpha)g_1^-(\zeta) + \alpha(s_L^{32}(\zeta) - \bar{\Delta}) = g_2^-(\zeta) > c.$$

Therefore all agents send message 3 at date 2. High-message consensus is absorbing because, if all agents send message 3, then every social aggregate is r , every current social-plus-private input is at least $r - \bar{\Delta} > c$, and any agent whose score is in $B_3 = (c, \infty)$ remains in B_3 . Thus all agents send message 3 from date 2 onward.

Finally fix $\zeta \in \mathcal{Z}_I$. Starting from $(3, 1)$, every H -agent and every L -agent satisfy, respectively,

$$-c < h_1^-(\zeta) \leq \ell_{i,1} \leq h_1^+(\zeta) \leq c, \quad -c < g_1^-(\zeta) \leq \ell_{i,1} \leq g_1^+(\zeta) \leq c.$$

Hence every agent sends message 2 at date 1, and the date-one block message profile is $(2, 2)$. Interior-message consensus is absorbing because, if all agents send message 2, then every social aggregate is 0, every current social-plus-private input lies in $[-\bar{\Delta}, \bar{\Delta}] \subset (-c, c)$, and any agent whose score lies in $B_2 = (-c, c]$ remains in B_2 . Thus all agents send message 2 from date 1 onward.

It remains to prove the ordering statement. The function $s_L^{31}(\zeta) = r(2q\zeta - 1)$ is strictly increasing in ζ . If $\zeta_S \in \mathcal{Z}_S$, then $s_L^{31}(\zeta_S) \leq -c - \bar{\Delta}$. If $\zeta_M \in \mathcal{Z}_M$, then $g_1^-(\zeta_M) > -c$. Since

$$g_1^-(\zeta_M) = (1 - \alpha)g_0 + \alpha(s_L^{31}(\zeta_M) - \bar{\Delta})$$

and $g_0 < -c$, it must be that $s_L^{31}(\zeta_M) - \bar{\Delta} > -c$; otherwise $g_1^-(\zeta_M)$ would be a convex combination of one number strictly below $-c$ and one number weakly below $-c$. Hence $s_L^{31}(\zeta_M) > -c + \bar{\Delta}$, so $s_L^{31}(\zeta_M) > s_L^{31}(\zeta_S)$. Strict monotonicity gives $\zeta_S < \zeta_M$.

Next,

$$h_1^-(\zeta) = (1 - \alpha)h_0 + \alpha(s_H^{31}(\zeta) - \bar{\Delta})$$

is strictly decreasing in ζ , because $s_H^{31}(\zeta) = r(1 - 2(1 - q)\zeta)$ is strictly decreasing. If $\zeta_M \in \mathcal{Z}_M$, then $h_1^-(\zeta_M) > c$. If $\zeta_I \in \mathcal{Z}_I$, then $h_1^-(\zeta_I) \leq h_1^+(\zeta_I) \leq c$. Therefore $h_1^-(\zeta_M) > h_1^-(\zeta_I)$, and strict monotonicity of h_1^- implies $\zeta_M < \zeta_I$. Combining the two inequalities gives $\zeta_S < \zeta_M < \zeta_I$. ■

Proof of Corollary 3

Fix $\zeta_M \in \mathcal{Z}_M$ and $\zeta_I \in \mathcal{Z}_I$. By Theorem 2, under ζ_M , every score is in $B_3 = (c, \infty)$ from date 2 onward. Hence, conditional on $\theta = 1$, every squared error is strictly below $(1 - \sigma(c))^2$, and therefore

$$\text{MSE}_t(1; \zeta_M) < (1 - \sigma(c))^2 \quad \text{for every } t \geq 2.$$

Under ζ_I , every score is in $B_2 = (-c, c]$ from date 1 onward. Thus every probability assessment is at most $\sigma(c)$, and every squared error is at least $(1 - \sigma(c))^2$. Therefore

$$\text{MSE}_t(1; \zeta_I) \geq (1 - \sigma(c))^2 \quad \text{for every } t \geq 2.$$

Combining the two inequalities gives

$$\text{MSE}_t(1; \zeta_M) < \text{MSE}_t(1; \zeta_I) \quad \text{for every } t \geq 2.$$

■

A.2 Auxiliary results

Lemma 3. *If $z \in [\underline{z}, \bar{z}]^n$, then*

$$\|\Pi z\|_n \leq \frac{\bar{z} - \underline{z}}{2}.$$

Proof of Lemma 3

The quantity $\|\Pi z\|_n^2$ is the empirical variance of z . The empirical variance of a vector whose entries lie in an interval of length $\bar{z} - \underline{z}$ is at most $(\bar{z} - \underline{z})^2/4$. Taking square roots gives the result. ■

Lemma 4. *Let $C \subseteq N$ satisfy $|C|/n \geq 1 - \varepsilon$. Then, for every $i \in N$,*

$$\sum_{j \notin C} P_{ij} \leq \omega(P)\varepsilon.$$

Proof of Lemma 4

Since $|N \setminus C| \leq \varepsilon n$,

$$\sum_{j \notin C} P_{ij} \leq |N \setminus C| \max_{a, b \in N} P_{ab} \leq \varepsilon n \max_{a, b \in N} P_{ab} = \omega(P)\varepsilon.$$

■

Proposition 2. Fix a set $C \subseteq N$, a message k , and $\xi \in [0, 1]$. Suppose

$$\sum_{j \notin C} P_{ij} \leq \xi \quad \text{for every } i \in C,$$

and suppose B_k is ξ -robust. If, at some date T , $\ell_{i,T} \in B_k$ for every $i \in C$, then, along every realization of future private increments satisfying $|\Delta_{i,s}| \leq \bar{\Delta}$,

$$\ell_{i,t} \in B_k \quad \text{and} \quad Q(\ell_{i,t}) = k$$

for every $i \in C$ and every $t \geq T$.

Proof of Proposition 2

The proof is by induction. The claim holds at date T by assumption. Suppose it holds at date $t \geq T$. Then every agent in C sends message k at date t . Fix $i \in C$, and write

$$\eta_i = \sum_{j \notin C} P_{ij}.$$

By assumption, $\eta_i \leq \xi$. Since every agent in C sends message k , the social aggregate can be written as

$$S_{i,t} = (1 - \eta_i)r_k + \sum_{j \notin C} P_{ij}\psi(\ell_{j,t}).$$

If $\eta_i > 0$, define

$$x_{i,t} = \frac{1}{\eta_i} \sum_{j \notin C} P_{ij}\psi(\ell_{j,t}).$$

Then $x_{i,t} \in [\underline{r}, \bar{r}]$, because all decoded values lie in $[\underline{r}, \bar{r}]$, and

$$S_{i,t} = (1 - \eta_i)r_k + \eta_i x_{i,t}.$$

If $\eta_i = 0$, the same representation holds after choosing any $x_{i,t} \in [\underline{r}, \bar{r}]$.

Because $r_k \in [\underline{r}, \bar{r}]$, $x_{i,t} \in [\underline{r}, \bar{r}]$, and $\eta_i \leq \xi$, we have

$$(1 - \xi)r_k + \xi\underline{r} \leq S_{i,t} \leq (1 - \xi)r_k + \xi\bar{r}.$$

Using $|\Delta_{i,t+1}| \leq \bar{\Delta}$, we obtain

$$\underline{h}_k(\xi) \leq S_{i,t} + \Delta_{i,t+1} \leq \bar{h}_k(\xi).$$

Since B_k is ξ -robust, the current input $S_{i,t} + \Delta_{i,t+1}$ lies in B_k . By the induction hypothesis, $\ell_{i,t} \in B_k$. The update

$$\ell_{i,t+1} = (1 - \alpha)\ell_{i,t} + \alpha(S_{i,t} + \Delta_{i,t+1})$$

is a convex combination of two points in $B_k = (b_{k-1}, b_k]$, and therefore also lies in B_k . Hence $Q(\ell_{i,t+1}) = k$. This completes the induction. $\blacksquare$

Lemma 5. *Maintain $\bar{\Delta} < c < r - \bar{\Delta}$. If all agents send message 3 at some date, then every agent who has a score in B_3 remains in B_3 at the next date. If all agents send message 2 at some date, then every agent who has a score in B_2 remains in B_2 at the next date. If all agents send message 1 at some date, then every agent who has a score in B_1 remains in B_1 at the next date.*

Proof of Lemma 5

If all agents send message 3, then $S_{i,t} = r$ for every i . Hence

$$S_{i,t} + \Delta_{i,t+1} \geq r - \bar{\Delta} > c.$$

If $\ell_{i,t} > c$, then the date- $t + 1$ score is a convex combination of two numbers above c , so $\ell_{i,t+1} > c$. Thus the agent remains in B_3 .

If all agents send message 1, then $S_{i,t} = -r$ for every i . Hence

$$S_{i,t} + \Delta_{i,t+1} \leq -r + \bar{\Delta} < -c.$$

If $\ell_{i,t} \leq -c$, then the date- $t + 1$ score is a convex combination of one number weakly below $-c$ and one number strictly below $-c$, so $\ell_{i,t+1} \leq -c$. Thus the agent remains in B_1 .

If all agents send message 2, then $S_{i,t} = 0$ for every i . Hence

$$-\bar{\Delta} \leq S_{i,t} + \Delta_{i,t+1} \leq \bar{\Delta}.$$

Since $\bar{\Delta} < c$, this current input lies in $(-c, c)$. A convex combination of a point in $B_2 = (-c, c]$ and a point in $(-c, c)$ remains in B_2 . ■

B Disagreement contraction and exposure mixing

This appendix records the sense in which the coefficient $\lambda(P)$ measures the speed of social mixing. Let P be row-stochastic.

Lemma 6. *For every $v \in \mathbb{R}^n$, $\|\Pi P v\|_n \leq \lambda(P) \|\Pi v\|_n$. Moreover, if $p_i^\top$ is row i of P and $\bar{p}^\top = \frac{1}{n} \mathbf{1}^\top P$, then*

$$(\Pi P v)_i = (p_i - \bar{p})^\top \Pi v \quad \text{for every } i.$$

Consequently, $\lambda(P) = 0$ if and only if all rows of P are identical.

Proof of Lemma 6

Since $P\mathbf{1} = \mathbf{1}$, we have $Pv = \bar{v}\mathbf{1} + P\Pi v$, and hence $\Pi P v = \Pi P \Pi v$. The first claim follows from the definition of $\lambda(P)$, since Πv is mean zero. The row formula follows from $\Pi P = P - \mathbf{1}\bar{p}^\top$ and from $(P - \mathbf{1}\bar{p}^\top)\mathbf{1} = 0$. If all rows of P are identical, then $\Pi P = 0$, so $\lambda(P) = 0$. Conversely, if $\lambda(P) = 0$, then $\Pi P v = 0$ for every mean-zero v . Hence $(p_i - \bar{p})^\top v = 0$ for every mean-zero v . Therefore $p_i - \bar{p}$ is proportional to $\mathbf{1}$. Since both p_i and $\bar{p}$ have row sum one, the proportionality constant is zero. Thus $p_i = \bar{p}$ for every i . ■

The row formula gives the exposure interpretation. Post-exposure disagreement is generated by differences between individual exposure rows and the average exposure row. If exposure rows are similar, then every mean-zero disagreement vector is averaged in a similar way by all agents, and $\lambda(P)$ is small. If exposure rows are segmented, some disagreement vectors survive exposure, and $\lambda(P)$ is large.

C Robustness for Block Mixing

This appendix collects auxiliary results for Section 5. First, we compute the projected mixing coefficient of the block matrix P^ζ , thereby connecting the block mixing parameter ζ to the general disagreement coefficient $\lambda(P)$. Second, we show that the same selection force arises when the lagging group starts from the interior message rather than the low message.

C.1 Projected mixing coefficient of the block matrix

Let $h = |H|$, $l = |L|$, and $q = h/n$. Recall that $h \geq 2$, $l \geq 2$, and $q \in (1/2, 1)$. The block matrix P^ζ is symmetric. Within each block all off-diagonal weights are equal, and across blocks

$$P_{ij}^\zeta = \frac{\zeta(1-q)}{|L|} = \frac{\zeta}{n} = \frac{\zeta q}{|H|} = P_{ji}^\zeta \quad (i \in H, j \in L).$$

Since P^ζ is symmetric and row-stochastic, it is doubly stochastic and self-adjoint under the empirical inner product

$$\langle u, v \rangle_n = \frac{1}{n} \sum_{i=1}^n u_i v_i.$$

Thus the projected mixing coefficient of P^ζ is the largest absolute eigenvalue of P^ζ on the mean-zero subspace.

Let

$$\lambda(P^\zeta) = \sup_{\bar{v}=0, v \neq 0} \frac{\|\Pi P^\zeta v\|_n}{\|v\|_n}, \quad \Pi = I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top.$$

Lemma 7. *For the block matrix P^ζ ,*

$$\lambda(P^\zeta) = \max \left\{ 1 - \zeta, \frac{1 - \zeta(1-q)}{|H| - 1}, \frac{1 - \zeta q}{|L| - 1} \right\}.$$

In particular, $\lambda(P^\zeta)$ is weakly decreasing in ζ .

Proof of Lemma 7

Because P^ζ is doubly stochastic, it maps the mean-zero subspace into itself. Hence, on that subspace, $\Pi P^\zeta = P^\zeta$. Because P^ζ is symmetric, its operator norm on the mean-zero subspace is the largest absolute eigenvalue of P^ζ on that subspace.

Decompose $\mathbb{R}^n$ into the mutually orthogonal subspaces generated by: $\mathbf{1}$, the block-contrast subspace

$$E_B = \{v : v_i = x \text{ for } i \in H, v_i = y \text{ for } i \in L, qx + (1-q)y = 0\},$$

the within- H subspace

$$E_H = \left\{ v : v_i = 0 \text{ for } i \in L, \sum_{i \in H} v_i = 0 \right\},$$

and the within- L subspace

$$E_L = \left\{ v : v_i = 0 \text{ for } i \in H, \sum_{i \in L} v_i = 0 \right\}.$$

These four subspaces are invariant under P^ζ .

The constant subspace has eigenvalue 1. This component is removed by the mean-zero projection.

Now take $v \in E_B$. Then $v_i = x$ on H , $v_i = y$ on L , and $qx + (1 - q)y = 0$. For $i \in H$,

$$(P^\zeta v)_i = (1 - \zeta(1 - q))x + \zeta(1 - q)y = x + \zeta(1 - q)(y - x).$$

Since $qx + (1 - q)y = 0$, we have $(1 - q)(y - x) = -x$. Therefore

$$(P^\zeta v)_i = (1 - \zeta)x.$$

Similarly, for $i \in L$,

$$(P^\zeta v)_i = \zeta qx + (1 - \zeta q)y = y + \zeta q(x - y),$$

and $q(x - y) = -y$. Therefore

$$(P^\zeta v)_i = (1 - \zeta)y.$$

Thus E_B is an eigenspace with eigenvalue $1 - \zeta$.

Next take $v \in E_H$. For $i \in H$,

$$(P^\zeta v)_i = \frac{1 - \zeta(1 - q)}{|H| - 1} \sum_{\substack{j \in H \\ j \neq i}} v_j = -\frac{1 - \zeta(1 - q)}{|H| - 1} v_i,$$

because $\sum_{j \in H} v_j = 0$. For $i \in L$, $(P^\zeta v)_i = 0$. Hence E_H is an eigenspace with eigenvalue

$$-\frac{1 - \zeta(1 - q)}{|H| - 1}.$$

The same argument on E_L gives eigenvalue

$$-\frac{1 - \zeta q}{|L| - 1}.$$

The mean-zero subspace is the orthogonal direct sum of E_B , E_H , and E_L . Therefore

$$\lambda(P^\zeta) = \max \left\{ 1 - \zeta, \frac{1 - \zeta(1 - q)}{|H| - 1}, \frac{1 - \zeta q}{|L| - 1} \right\}.$$

Each term inside the maximum is weakly decreasing in ζ , so $\lambda(P^\zeta)$ is weakly decreasing in ζ . ■

Lemma 7 gives the formal connection between the block parameter ζ and the general mixing measure used in Section 4. The term $1 - \zeta$ is the block-level contraction coefficient. The remaining two terms are within-block residual coefficients generated by the no-self-weight convention. These residual terms also decline with ζ , and they vanish as the corresponding block sizes grow.

C.2 Robustness to the initial profile (3, 2)

The main block result starts from the initial profile (3, 1): group H sends the state-aligned message and group L sends the state-opposed message. We now show that the same selection force does not require the lagging group to begin at the bottom of the codebook. It is enough that group H has crossed into the state-aligned bin while group L has not.

Assume

$$\ell_{i,0} = h_0 > c \quad \text{for } i \in H, \quad \ell_{i,0} = m_0 \in (-c, c) \quad \text{for } i \in L.$$

The initial message profile is therefore (3, 2). The block social inputs generated by this profile are

$$s_H^{32}(\zeta) = r(1 - \zeta(1 - q)), \quad s_L^{32}(\zeta) = \zeta qr.$$

The first expression is strictly decreasing in ζ , while the second is strictly increasing in ζ .

After one update from the profile (3, 2), the possible date-one scores are contained in the intervals

$$I_H^{32}(\zeta) = [\underline{h}_1^{32}(\zeta), \bar{h}_1^{32}(\zeta)], \quad I_L^{32}(\zeta) = [\underline{m}_1^{32}(\zeta), \bar{m}_1^{32}(\zeta)],$$

where

$$\underline{h}_1^{32}(\zeta) = (1 - \alpha)h_0 + \alpha(s_H^{32}(\zeta) - \bar{\Delta}), \quad \bar{h}_1^{32}(\zeta) = (1 - \alpha)h_0 + \alpha(s_H^{32}(\zeta) + \bar{\Delta}),$$

and

$$\underline{m}_1^{32}(\zeta) = (1 - \alpha)m_0 + \alpha(s_L^{32}(\zeta) - \bar{\Delta}), \quad \bar{m}_1^{32}(\zeta) = (1 - \alpha)m_0 + \alpha(s_L^{32}(\zeta) + \bar{\Delta}).$$

Define the asymmetric, high-consensus, and interior-consensus sufficient regions by

$$\mathcal{Z}_A^{32} = \{\zeta \in [0, 1] : s_H^{32}(\zeta) - \bar{\Delta} > c, s_L^{32}(\zeta) - \bar{\Delta} > -c, s_L^{32}(\zeta) + \bar{\Delta} \leq c\},$$

$$\mathcal{Z}_H^{32} = \{\zeta \in [0, 1] : I_H^{32}(\zeta) \subset (c, \infty), I_L^{32}(\zeta) \subset (c, \infty)\},$$

and

$$\mathcal{Z}_I^{32} = \{\zeta \in [0, 1] : I_H^{32}(\zeta) \subset (-c, c], I_L^{32}(\zeta) \subset (-c, c]\}.$$

Proposition 3. *Starting from the initial profile $(3, 2)$, the following statements hold pathwise along every realization satisfying the private-increment bound.*

1. *If $\zeta \in \mathcal{Z}_A^{32}$, then $(3, 2)$ is absorbing. Group H sends message 3 forever, and group L sends message 2 forever.*
2. *If $\zeta \in \mathcal{Z}_H^{32}$, then the path $(3, 2) \rightarrow (3, 3)$ is selected. All agents send message 3 from date 1 onward.*
3. *If $\zeta \in \mathcal{Z}_I^{32}$, then the path $(3, 2) \rightarrow (2, 2)$ is selected. All agents send message 2 from date 1 onward.*
4. *The asymmetric region is ordered below either consensus region. If $\zeta_A \in \mathcal{Z}_A^{32}$ and $\zeta_H \in \mathcal{Z}_H^{32}$, then $\zeta_A < \zeta_H$. If $\zeta_A \in \mathcal{Z}_A^{32}$ and $\zeta_I \in \mathcal{Z}_I^{32}$, then $\zeta_A < \zeta_I$.*
5. *The two consensus regions cannot both be nonempty under the same primitives:*

$$\mathcal{Z}_H^{32} \neq \emptyset \implies \mathcal{Z}_I^{32} = \emptyset,$$

and

$$\mathcal{Z}_I^{32} \neq \emptyset \implies \mathcal{Z}_H^{32} = \emptyset.$$

Proof of Proposition 3

We first prove the pathwise conclusions.

If $\zeta \in \mathcal{Z}_A^{32}$, then under the profile $(3, 2)$, every H -agent's current social-plus-private input lies above c , because $s_H^{32}(\zeta) - \bar{\Delta} > c$. Every L -agent's current social-plus-private input lies in $(-c, c]$, because

$$s_L^{32}(\zeta) - \bar{\Delta} > -c \quad \text{and} \quad s_L^{32}(\zeta) + \bar{\Delta} \leq c.$$

Since H -agents start in $B_3 = (c, \infty)$ and L -agents start in $B_2 = (-c, c]$, convexity of the update rule implies that all H -agents remain in B_3 and all L -agents remain in B_2 . The same argument applies at every later date, so $(3, 2)$ is absorbing.

If $\zeta \in \mathcal{Z}_H^{32}$, then both date-one score intervals are contained in B_3 . Hence all agents send message 3 from date 1 onward. Once all agents send message 3, every social input is r . Since $r - \bar{\Delta} > c$, high-message consensus is absorbing.

If $\zeta \in \mathcal{Z}_I^{32}$, then both date-one score intervals are contained in B_2 . Hence all agents send message 2 from date 1 onward. Once all agents send message 2, every social input is 0. Since $\bar{\Delta} < c$, every current social-plus-private input lies in $(-c, c)$, so interior-message consensus is absorbing.

We now prove the ordering claims. The two block inputs satisfy

$$s_H^{32}(\zeta) = r(1 - \zeta(1 - q)), \quad s_L^{32}(\zeta) = \zeta qr.$$

Thus s_H^{32} is strictly decreasing in ζ , while s_L^{32} is strictly increasing in ζ .

Take $\zeta_A \in \mathcal{Z}_A^{32}$ and $\zeta_H \in \mathcal{Z}_H^{32}$. Since $\zeta_A \in \mathcal{Z}_A^{32}$, $s_L^{32}(\zeta_A) + \bar{\Delta} \leq c$. Since $\zeta_H \in \mathcal{Z}_H^{32}$, $(1 - \alpha)m_0 + \alpha(s_L^{32}(\zeta_H) - \bar{\Delta}) > c$.

Because $m_0 < c$, this implies $s_L^{32}(\zeta_H) - \bar{\Delta} > c$. Therefore

$$s_L^{32}(\zeta_A) \leq c - \bar{\Delta} < c + \bar{\Delta} < s_L^{32}(\zeta_H).$$

Since s_L^{32} is strictly increasing, $\zeta_A < \zeta_H$.

Next take $\zeta_A \in \mathcal{Z}_A^{32}$ and $\zeta_I \in \mathcal{Z}_I^{32}$. Since $\zeta_A \in \mathcal{Z}_A^{32}$, $s_H^{32}(\zeta_A) - \bar{\Delta} > c$. Since $\zeta_I \in \mathcal{Z}_I^{32}$, $(1 - \alpha)h_0 + \alpha(s_H^{32}(\zeta_I) + \bar{\Delta}) \leq c$.

Because $h_0 > c$, this implies $s_H^{32}(\zeta_I) + \bar{\Delta} < c$. Therefore $s_H^{32}(\zeta_A) > c + \bar{\Delta} > c - \bar{\Delta} > s_H^{32}(\zeta_I)$. Since s_H^{32} is strictly decreasing, $\zeta_A < \zeta_I$.

Finally, suppose toward a contradiction that both consensus regions are nonempty. Choose

$$\zeta_H \in \mathcal{Z}_H^{32} \quad \text{and} \quad \zeta_I \in \mathcal{Z}_I^{32}.$$

The H -block conditions imply $(1 - \alpha)h_0 + \alpha(s_H^{32}(\zeta_H) - \bar{\Delta}) > c$ and $(1 - \alpha)h_0 + \alpha(s_H^{32}(\zeta_I) + \bar{\Delta}) \leq c$. Hence $s_H^{32}(\zeta_H) > s_H^{32}(\zeta_I)$. Because s_H^{32} is strictly decreasing, this implies $\zeta_H < \zeta_I$.

The L -block conditions imply

$$(1 - \alpha)m_0 + \alpha(s_L^{32}(\zeta_H) - \bar{\Delta}) > c$$

and

$$(1 - \alpha)m_0 + \alpha(s_L^{32}(\zeta_I) + \bar{\Delta}) \leq c.$$

Hence $s_L^{32}(\zeta_H) > s_L^{32}(\zeta_I)$. Because s_L^{32} is strictly increasing, this implies $\zeta_H > \zeta_I$. This contradiction proves that $\mathcal{Z}_H^{32}$ and $\mathcal{Z}_I^{32}$ cannot both be nonempty. ■

Proposition 3 shows that the main selection force does not depend on group L starting from the state-opposed message. Starting from $(3, 2)$, increasing ζ raises the lagging group’s exposure to the state-aligned message, but it also lowers the initially state-aligned group’s exposure to its own message. The same tradeoff therefore remains: more mixing can move the lagging group upward, but excessive mixing can pull the leading group into the interior before high consensus forms.

References

- Acemoglu, D. et al. (2011). “Bayesian Learning in Social Networks”. *The Review of Economic Studies* 78.4, pp. 1201–1236.
- Amir, G. et al. (2025). “Granular DeGroot Dynamics: A Model for Robust Naive Learning in Social Networks”. *Journal of Economic Theory* 223, p. 105952.
- Banerjee, A. V. (1992). “A Simple Model of Herd Behavior”. *The Quarterly Journal of Economics* 107.3, pp. 797–817.
- Bikhchandani, S., D. Hirshleifer, and I. Welch (1992). “A Theory of Fads, Fashion, Custom, and Cultural Change as Informational Cascades”. *Journal of Political Economy* 100.5, pp. 992–1026.
- Currarini, S., M. O. Jackson, and P. Pin (2009). “An Economic Model of Friendship: Homophily, Minorities, and Segregation”. *Econometrica* 77.4, pp. 1003–1045.
- DeGroot, M. H. (1974). “Reaching a Consensus”. *Journal of the American Statistical Association* 69.345, pp. 118–121.
- DeMarzo, P. M., D. Vayanos, and J. Zwiebel (2003). “Persuasion Bias, Social Influence, and Unidimensional Opinions”. *The Quarterly Journal of Economics* 118.3, pp. 909–968.
- Golub, B. and M. O. Jackson (2010). “Naive Learning in Social Networks and the Wisdom of Crowds”. *American Economic Journal: Microeconomics* 2.1, pp. 112–149.
- Golub, B. and M. O. Jackson (2012). “How Homophily Affects the Speed of Learning and Best-Response Dynamics”. *The Quarterly Journal of Economics* 127.3, pp. 1287–1338.
- Holland, P. W., K. B. Laskey, and S. Leinhardt (1983). “Stochastic Blockmodels: First Steps”. *Social Networks* 5.2, pp. 109–137.
- Jadbabaie, A. et al. (2012). “Non-Bayesian Social Learning”. *Games and Economic Behavior* 76.1, pp. 210–225.
- Kashyap, A., T. Başar, and R. Srikant (2007). “Quantized Consensus”. *Automatica* 43.7, pp. 1192–1203.

- Lizzeri, A. (1999). “Information Revelation and Certification Intermediaries”. *The RAND Journal of Economics* 30.2, pp. 214–231.
- Molavi, P., A. Tahbaz-Salehi, and A. Jadbabaie (2018). “A Theory of Non-Bayesian Social Learning”. *Econometrica* 86.2, pp. 445–490.
- Onuchic, P. and D. Ray (2023). “Conveying Value via Categories”. *Theoretical Economics* 18.4, pp. 1407–1439.
- Smith, L. and P. Sørensen (2000). “Pathological Outcomes of Observational Learning”. *Econometrica* 68.2, pp. 371–398.
- Xu, W. (2025). “Social Learning through Coarse Signals of Others’ Actions”. *Journal of Economic Theory* 229, p. 106066.